YEAR 7 · UNIT 4 · HOMEWORK 9

Lighter on the Moon

Mathematics 4.1–4.5 and the Unit 4 review — decimals
Section F — Science 3.1 in decimals: weight, gravity and falling

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 23 October 2026

How to do this homework

Six sections. A to E are the decimal lessons. F is the science from 3.1, and **every answer in F is a decimal**.

- **Write the calculation before you write the answer.**
- **Estimate first.** The estimate tells you where the point goes.
- **No calculator.** In F, answer in **full English sentences** where it says so.

Section A · Mathematics 4.1 — Comparing and Ordering

A1 Put each number on the line

The line goes from **5** to **6**. Every small line is **0.1**. **(a)** Draw an arrow to each number and write its letter.



A 5.2	**B** 5.9	**C** 5.45	**D** 5.05

(b) Now write the four numbers **in order of size, starting with the smallest**.

_____

A2 Which is bigger?

Compare decimals

First look at the **whole-number parts**. If they are the same, **fill with zeros** so both numbers have the same number of digits after the point, then compare. 3.3 = 3.30, and 3.30 > 3.25.

Write <, > or = in each box.

(a) 7.4 ☐ 7.38 (c) 3.05 ☐ 3.5 (e) 0.70 ☐ 0.7

(b) 0.6 ☐ 0.59 (d) 12.3 ☐ 9.99 (f) 4.125 ☐ 4.13

(g) Mr Bowen says: *"6.18 is bigger than 6.2, because 18 is bigger than 2."*

(i) Write how you **say** each number in English.

6.18: _____ 6.2: _____

(ii) Is he right? Explain. Start: **"Mr Bowen is … because …"**

A3 Who won?

Word help — the words that tell you which end

longest the biggest distance

fastest the **smallest** time. Fast runners and swimmers take **less** time

fourth number 4 in the order. *Fourth fastest*: the fourth smallest time

Long jump

Jumper	Distance (m)
A	4.8
B	4.08
C	4.75
D	4.79
E	4.5

50 m swim

Lane	Time (s)
1	31.4
2	30.95
3	31.09
4	30.9
5	31.12

(a) Write the five jumps in order of size, starting with the smallest.

_____

(b) The **longest** jump wins. Which jumper won? _____

(c) Write the five swim times in order of size, starting with the smallest.

_____

(d) Which lane was the **fastest**? _____

(e) Which lane came **fourth**? _____

(f) Which lane was the **slowest**? _____

Section B · Mathematics 4.2 — Adding and Subtracting

B1 Use a written method

Line up the points

A **written method** is in columns on paper. Write the numbers with the **points in a line**. Fill the gaps with **zeros**. Work from the right, like whole numbers. The point in the answer goes **straight down**.

(a) 5.36 + 2.7

(b) 14.8 + 3.65

(c) 0.45 + 6 + 1.9

(d) 9.4 − 3.75

(e) 12 − 4.36

(f) 1 − 0.65

B2 Use a mental method

A **mental method** is in your head. Change one number to a whole number, then give back the difference. Write the easier calculation, then the answer. The first one is done for you.

Calculation	The easier way	Answer
5.8 + 2.9	5.8 + 3 − 0.1	8.7
3.9 + 2.5		
7.3 − 1.9		
6.99 + 3.45		
12.6 − 4.98		

Section C · Mathematics 4.3–4.4 — Multiplying and Dividing

C1 Point out, point back

Multiplying a decimal

Point out: 32 × 7 = 224. Point back, **same number of decimal places**: 3.2 × 7 = 22.4
Estimate to check: 3 × 7 = 21, and 22.4 is close to 21.

Fill in the table. The first one is done for you.

Calculation	Estimate	Without the point	Answer
4.6×3	$5 \times 3 = 15$	$46 \times 3 = 138$	13.8
6.4×5			
0.7×8			
2.35×4			
8.15×2			

C2 The point goes straight up

Dividing a decimal

Use short division. The point in the answer goes **straight up**.
If a digit is too small to divide, write **0** and carry it.
Check with the **inverse calculation**: $8.24 \div 4 = 2.06$ because $2.06 \times 4 = 8.24$.

(a) $7.56 \div 3$

(b) $6.24 \div 6$

(c) $0.84 \div 7$

(d) $5.4 \div 4$

(e) $16.2 \div 5$

(f) $8.08 \div 8$

(g) Check **(b)** with the inverse calculation. _____ × _____ = _____

C3 One fact, five answers

You are given one fact. **Do not multiply again.** Use the fact, and an estimate to decide where the point goes.

$38 \times 27 = 1026$

(a) $3.8 \times 27 =$ _____

(b) $0.38 \times 27 =$ _____

(c) $38 \times 2.7 =$ _____

(d) $102.6 \div 27 =$ _____

(e) $10.26 \div 27 =$ _____

(f) Two of your answers are the same. Which two, and why?

Section D · Mathematics 4.5 — Making Calculations Easier

D1 Decimals as fractions

Word help

equivalent	equal in value, but written differently. 0.3 and $\frac{3}{10}$ are equivalent
tenths · hundredths	the column of the **last** digit is the bottom number

Write each decimal as a fraction, and each fraction as a decimal.

(a) 0.7 = _____ (c) 0.31 = _____ (e) $\frac{8}{100}$ = _____

(b) 0.03 = _____ (d) 0.6 = _____ (f) $\frac{51}{100}$ = _____

D2 Half of, a quarter of, a tenth of

In English, "of" often means ×

× 0.5 = **half of** = ÷ 2 × 0.25 = **a quarter of** = ÷ 4 × 0.1 = **a tenth of** = ÷ 10

Fill in the table. The first one is done for you.

Calculation	In English	Answer
48 × 0.5	half of 48	24
36 × 0.25		
70 × 0.1		
	a quarter of 2.8	
	a tenth of 4.5	

D3 Multiplying by a decimal

Divide first, then multiply

× 0.2 means ÷ 10, then × 2. × 0.03 means ÷ 100, then × 3.
So 250 × 0.2 = 25 × 2 = 50. Multiplying by a decimal **less than 1** makes a number **smaller**.

(a) 40 × 0.3 = _____ × _____ = _____ (c) 600 × 0.04 = _____ × _____ = _____

(b) 150 × 0.2 = _____ × _____ = _____ (d) 90 × 0.7 = _____ × _____ = _____

(e) $1200 \times 0.05 =$ _____ × _____ = _____ (f) $80 \times 0.6 =$ _____ × _____ = _____

D4 Times 9, times 11, and partitioning

Three quick ways

× 9: do × 10, then **take away** one lot. $3.8 \times 9 = 38 - 3.8 = 34.2$
× 11: do × 10, then **add** one lot. $3.8 \times 11 = 38 + 3.8 = 41.8$
Partitioning: split the number. $4.3 \times 6 = 4 \times 6 + 0.3 \times 6 = 24 + 1.8 = 25.8$

Write the quick way, then the answer.

(a) $4.6 \times 9 =$ _____ = _____

(b) $2.3 \times 11 =$ _____ = _____

(c) $6.2 \times 7 =$ _____ = _____

(d) $3.8 \times 5 =$ _____ = _____

D5 Mr Bowen's homework

Mr Bowen gave himself 4 out of 4. Only **one** of these is right. Find it, and correct the other three.

(a) $0.4 \times 7 = 0.28$ _____

(b) $5.09 > 5.1$ _____

(c) $7.2 \div 6 = 1.2$ _____

(d) $50 \times 0.3 = 150$ _____

Section E · Unit 4 Review — Bigger Numbers, Real Problems

E1 Two-digit numbers

Estimate, or write the table

× a 2-digit number: take the point out, multiply, then use an **estimate** to put it back.
4.7×32: $47 \times 32 = 1504$, and $5 \times 30 = 150$, so **150.4**.
÷ a 2-digit number: write its **times table** first, then use short division. The point goes straight up.

(a) 5.8×15

(b) $41.6 \div 13$

E2 From English to a calculation

Word help — the words that choose the operation

change	in a shop, the money you get **back** when you pay too much. Take away
share equally	give everyone the same amount. Divide — and count the people carefully
how much heavier	the difference. Take away
subtract 2 from 9	9 − 2. The **second** number comes **first**

Write the **calculation first**, then the answer. Give units.

(a) Mr Bowen pays for a 3.65-dollar sandwich with a 5-dollar note. How much change does he get?

Calculation: _____ **Answer:** _____

(b) A mango costs 1.20 dollars. How much do 6 mangoes cost?

Calculation: _____ **Answer:** _____

(c) Mr Bowen and two friends share a 13.50-dollar pizza equally. How much does each person pay?

Calculation: _____ **Answer:** _____

(d) A bag of rice weighs 2.4 kg. A bag of sugar weighs 0.85 kg. How much heavier is the rice?

Calculation: _____ **Answer:** _____

(e) Subtract 2.6 from 9.1.

Calculation: _____ **Answer:** _____

E3 Money has two decimal places

Money

Money is written to **2 decimal places**. 4.5 dollars is written **4.50** dollars.
10 ÷ 3 = 3.333 . . ., so three friends get **3.33** dollars each, and 1 cent is left over.

Write each one as money.

(a) 4.8 dollars _____ **(b)** 9.125 dollars _____ **(c)** 15 ÷ 4 dollars _____

(d) Three friends share 25 dollars equally. How much does each friend get? How many cents are left over?

Each friend: _____ dollars Left over: _____ cents

E4 Word problems

(a) Mr Bowen buys 3 notebooks at 2.45 dollars each and one pen for 1.80 dollars. He pays with a 10-dollar note. How much change does he get?

(b) Mr Bowen buys 0.5 kg of grapes. He eats a quarter of them on the walk home, which he calls "quality control". How many kilograms of grapes arrive at his house?

E5 Now you write the question

The rules

A **real** situation — a shop, a recipe, a race, a bag on a scale. Full sentences, ending in a **question**. Use one of the **words** from E2. Give **units**. Your question must come out at the answer given.

Write a word problem for $12.60 \div 4 = 3.15$

Section F · Science 3.1 in decimals — Mr Bowen in Space

Weight and mass, from Science 3.1

weight (N) = mass (kg) × strength of gravity (N/kg)
On Earth, the strength of gravity is **10 N/kg**. Say it: **"ten newtons per kilogram"**.
Mass does not change from planet to planet. Weight does.

F1 How strong is gravity?

These are the strengths of gravity from the lesson. A bigger number pulls harder.

Place	Moon	Mars	Venus	Uranus	Earth	Neptune	Jupiter
N/kg	**1.6**	**3.7**	**8.9**	**8.7**	**10**	**11**	**25**

(a) Write the seven strengths in order of size, starting with the smallest.

_____

(b) Which pulls harder, **Venus** or **Uranus**? By how much? _____

(c) How much stronger is gravity on the **Earth** than on the **Moon**? _____

F2 Mr Bowen's space backpack

Mr Bowen's space backpack has a mass of **15 kg**. **(a)** Find its weight in each place. Estimate first. The first one is done for you.

Place	Gravity (N/kg)	Calculation	Weight (N)
Earth	10	15×10	150
Moon	1.6		
Mars	3.7		
Venus	8.9		
Jupiter	25		

(b) How much **less** does the backpack weigh on the Moon than on the Earth? _____

F3 Do heavy things fall faster?

Mr Bowen drops a **heavy** ball and a **light** ball, the same size, from 1 metre. He times each fall with a stopwatch, four times.

	Drop 1	**Drop 2**	**Drop 3**	**Drop 4**
Heavy ball (s)	0.48	0.41	0.50	0.41
Light ball (s)	0.44	0.52	0.39	0.45

Word help

conjecture	a guess that you can test
mean	add up all the values, then divide by how many there are
reaction time	how long you take to move. It is about **0.2 s**

(a) Find the **fastest** time in the table. Which ball was it? _____

(b) Make a **conjecture** from (a): which ball falls faster? _____

(c) Now test it. Find the **mean** time for each ball.

Heavy ball **Light ball**

(d) Do heavy things fall faster? Use your two means. Answer in a full sentence.

(e) The four times for the **same** ball are all different. Why? Use the words **reaction time**.

F4 Mr Bowen moves house

Mr Bowen's mass is **80 kg**. He would like to weigh exactly **296 N** without eating any less. Which place in **F1** should he move to? Show your calculation.

He has started packing. What will his **mass** be when he gets there? _____